

TABLE XIII: **Ablation study on auxiliary loss weights.** All runs use 30k training steps and a global batch size of 64; only the object and skill supervision weights vary. When a configuration does not explicitly vary one of the weights, that weight is set to 0.01. We report the Light, Background, and Layout tracks, and Avg denotes their arithmetic mean. Our final setting uses balanced low weights ($w_{\text{obj}}, w_{\text{skill}} = (0.001, 0.001)$) and achieves the best average score.

Setting	w_{obj}	w_{skill}	Light	Background	Layout	Avg
Final (Ours)	0.001	0.001	94.60	89.00	79.90	87.83
High obj. / high skill	0.01	0.01	92.05	88.07	77.19	85.77
High obj. / low skill	0.01	0.001	92.46	89.42	76.48	86.12
Low obj. / high skill	0.001	0.01	91.51	88.56	75.59	85.22
Low obj. / lower skill	0.001	0.0003	91.17	89.96	76.00	85.71

laboratory manipulation tasks on the PSI-Bot dual-arm platform (T4–T6). Each evaluation trial lasts for at most **120 s** and terminates early once the success condition is met. Unless otherwise specified, we require a **1 s dwell time**: the relevant objects must remain stable in the target configuration for at least 1 s without human intervention.

ALOHA household tasks:

(T1) Pick up fruits and vegetables. The robot must place the green pepper and carrot onto the plate, and place the strawberry into the bowl. A trial is successful if the pepper and carrot are both inside the plate region and the strawberry is inside the bowl region.

(T2) Stack bowls and place on the first shelf. The robot stacks two bowls and places the stacked bowls onto the first shelf. A trial is successful if the bowls form a stable stacked configuration and the stack is placed within the designated shelf region.

(T3) Clean the tabletop (sweep → dustpan → pour → return). The robot sweeps trash into the dustpan with a broom, pours the trash from the dustpan into the tray, and returns both the broom and dustpan back to the table. A trial is successful if the robot completes the pouring action over the tray and returns the tools to the table.

PSI-Bot laboratory tasks:

(T4) Place beaker in heating mantle. The robot grasps a beaker and inserts it into the heating mantle. A trial is successful if the beaker bottom is seated inside the mantle opening (i.e., inserted into the cavity).

(T5) Stack small beakers inside a large beaker. The robot places small beakers into a large beaker. A trial is successful if the small beakers are contained within the large beaker.

(T6) Heat the beaker (place the asbestos mesh, then place the beaker on it). The robot first places the asbestos mesh on the lower level of the iron stand, and then places the beaker on top of the mesh. A trial is successful if the mesh is placed on the designated lower support ring and the beaker is stably placed on the mesh.

M. Real-World Generalization Settings

We evaluate three real-robot generalization regimes: **in-domain (positional)**, **scene**, and **lighting**. Each trial is first reset to a canonical task layout and then randomized according

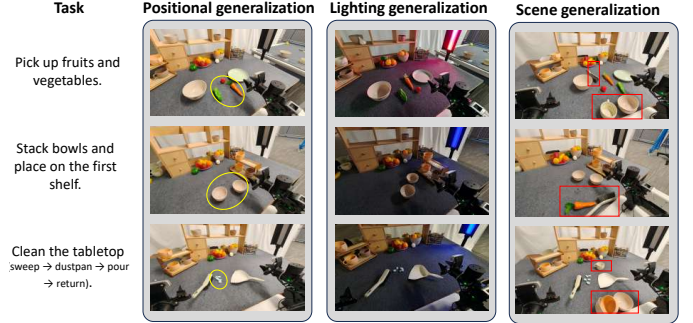


Fig. 11: **ALOHA real-world generalization settings (T1–T3).** From left to right: **in-domain (positional)** perturbations using a 3×3 anchor grid, **lighting** shifts with colored illumination, and **scene** shifts by adding distractor objects.

to *exactly one* regime; we do not combine multiple shifts within a single trial.

a) *In-domain (positional) generalization.*: We perturb the initial object placement within the training distribution by sampling from a 3×3 grid of 9 discrete anchors centered at the nominal pose. Adjacent anchors are spaced by 1–2 cm (approximately within ± 2 cm per axis relative to the nominal position), while keeping task semantics unchanged.

b) *Scene generalization.*: We introduce clutter by adding 3–5 distractor objects per trial, sampled from the same domain as the task (household items for ALOHA tasks, lab items for PSI-Bot tasks). Distractors are placed to avoid occluding target objects and to keep the nominal manipulation corridor feasible, thereby inducing appearance/context shifts without altering the intended task.

c) *Lighting generalization.*: We change illumination using colored decorative lighting with three color settings. Lighting is kept constant within each trial and constrained not to render target objects visually ambiguous, inducing appearance shifts while preserving task observability.

N. Visualization of LIBERO-Plus Results

To complement the quantitative success rates on simulation-based benchmarks in Table IV, we visualize representative *successful* policy rollouts from the LIBERO-Plus benchmark, covering its four task suites: **spatial**, **object**, **goal**, and **long**. Each visualization shows a sequence of 7 keyframes sampled